COMPUTER SYSTEMS AND PLATFORM TECHNOLOGIES

IT0047

FEU ALABANG FEU DILIMAN FEU TECH

Technology Driven by Innovation

MODULE 4

The System Bus, Von Neuman Architecture & CPU Registers

At the end of the lesson, the student should be able to :

- Define Bus
- Identify types of Bus
- Identify the Memory Operations

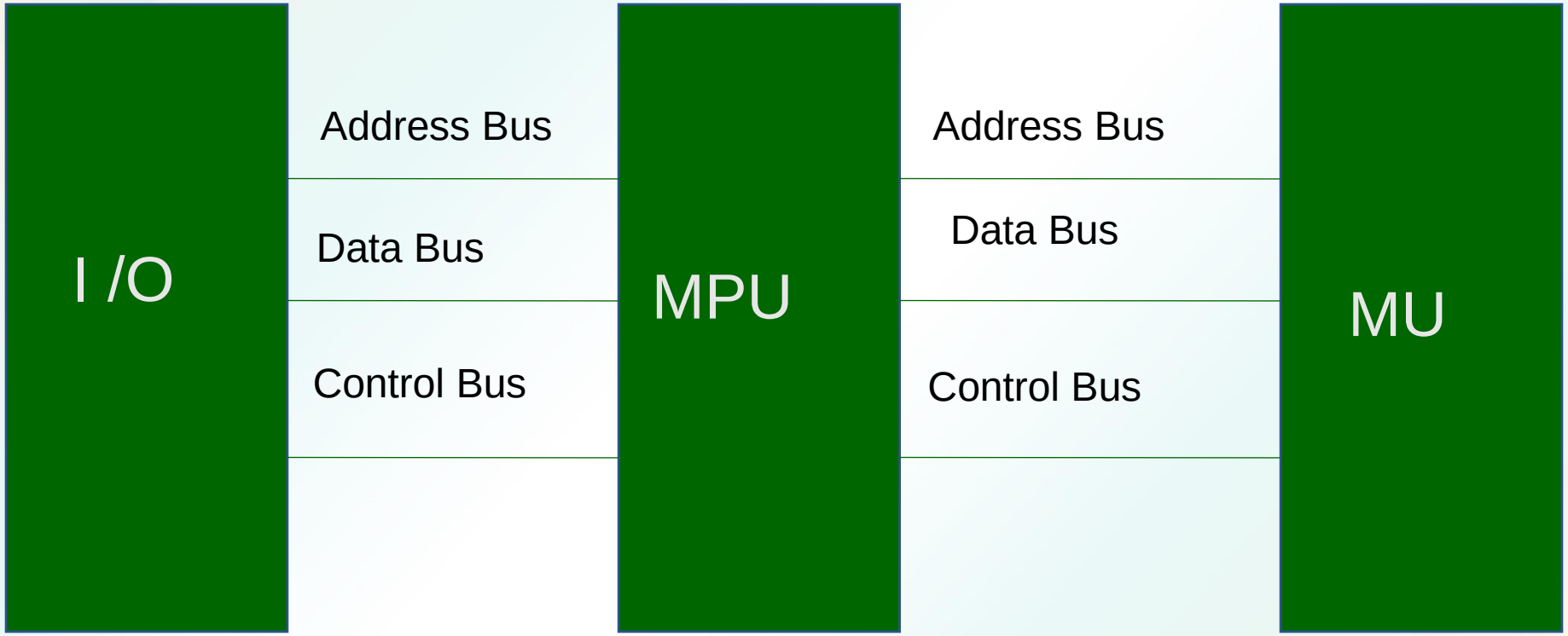
MODULE 4.2

Basic Operational Concept

At the end of the lesson, the student should be able to :

- **Familiar with the Von Neumann Architecture**
- **Identify the functions of the different components of the Von Neumann Architecture**
- **Identify the operating steps of the Von Neumann Architecture**

Von Neumann Architecture

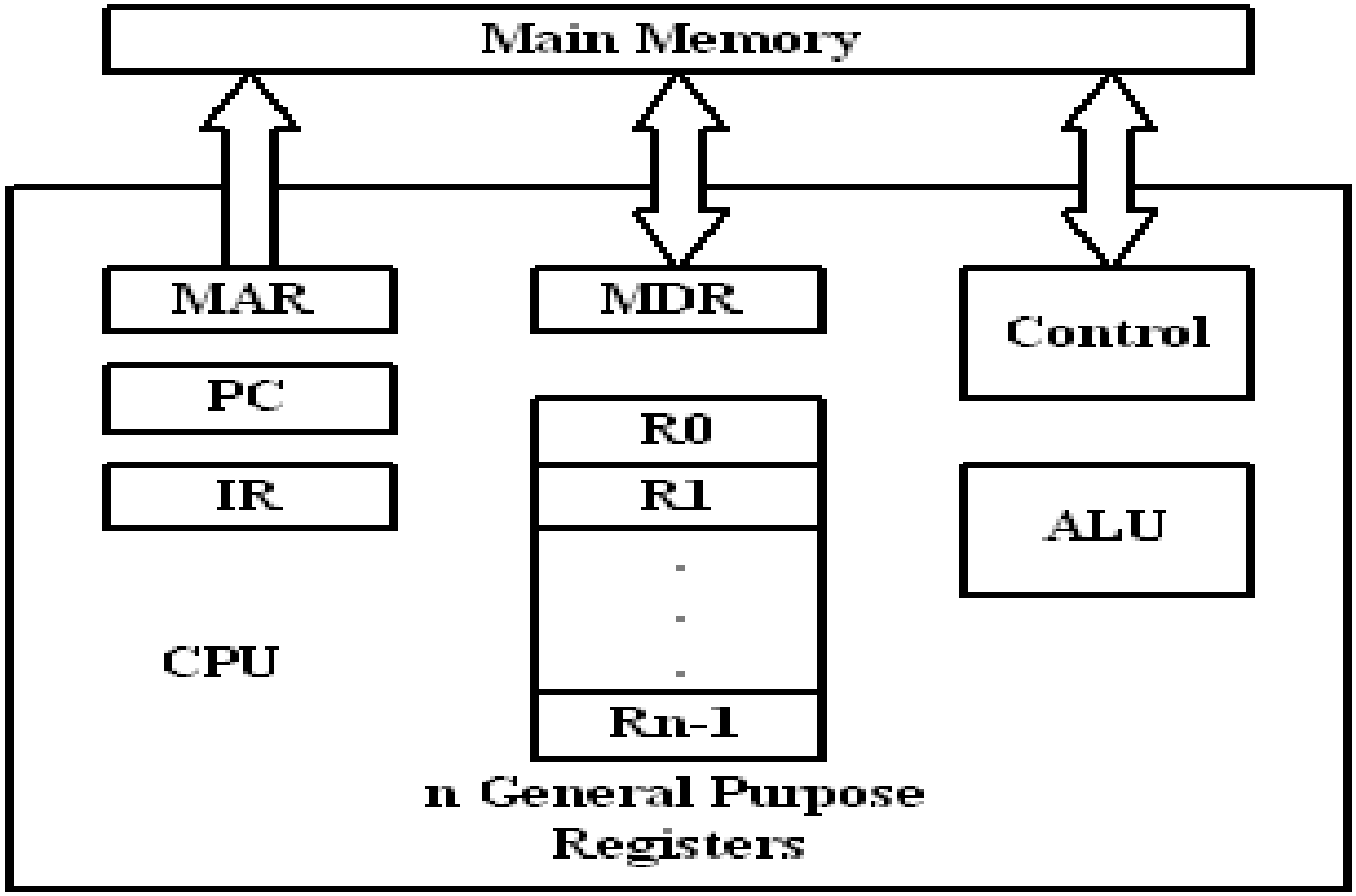


Von Neumann Architecture

Von Neumann Architecture means that programs (together with data) are stored in main memory during execution.

It is also known as the ***Stored Program Architecture*** or the ***Fetch-Decode-Execute Architecture***.

Connections between the processor and the main memory:



Functions of the Components

The PC (***Program Counter***) contains the memory address of the instruction to be executed. During execution, the contents of the PC are updated to point to the next instruction.

The MAR (***Memory Address Register***) holds the address of the location to or from which data are to be transferred.

The MDR (***Memory Data Register***) contains the data to be written or read out of the addressed location.

The IR (***Instruction Register***) contains the instruction that is being executed.

Operating Steps:

1. PC is set to point to the first instruction of the program (the operating system loads the memory address of the first instruction).
2. The contents of the PC are transferred to the MAR (which are automatically transmitted to the MM) and a *Read* signal is sent to the MM.
3. The addressed word is read out of MM and loaded into the MDR.
4. The contents of MDR are transferred to the IR. The instruction is ready to be decoded and executed.
5. During execution, the contents of the PC are incremented or updated to point to the next instruction.

Things to remember:

If operand or data needed by the instruction resides in MM:

1. It will have to be fetched by sending its address to the MAR and initiating a read cycle.

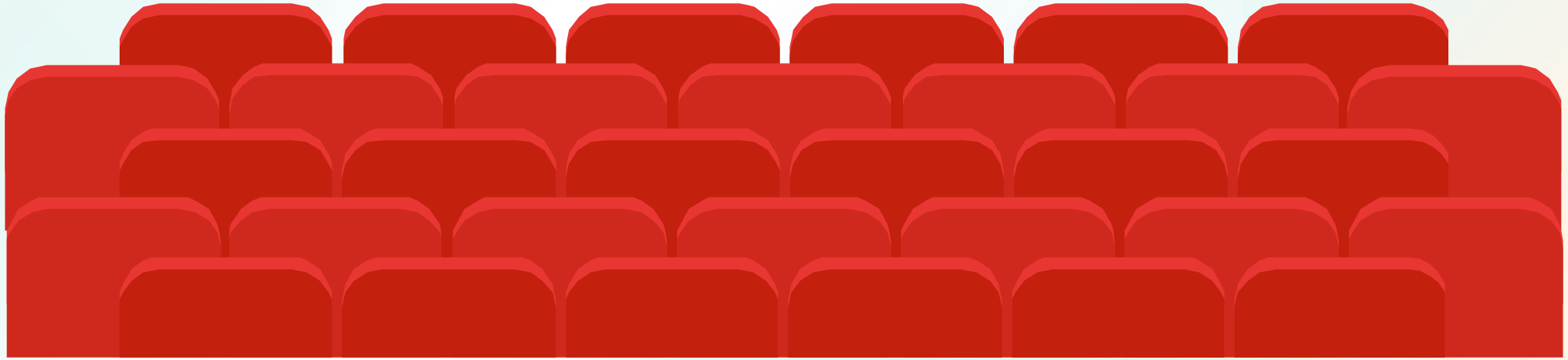
2. When the operand has been read from MM into the MDR, it may be transferred from the MDR to the ALU.

If result is to be stored in MM:

1. The result is sent to the MDR.

2. The address of the location where the result is to be stored is sent to the MAR and a write cycle is initiated

Q&A SESSION

ASK ANY QUESTION RELATED TO OUR TOPIC FOR TODAY.

THANK YOU